

MAYCOL ALEJANDRO ZARAZA AGUILERA

Earth Observation & Geospatial AI Specialist

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PROFESSIONAL PROFILE

Cadastral and Geodesist Engineer and geospatial specialist with more than eight years of professional experience in Earth Observation, Remote Sensing, Geographic Information Systems, Geospatial AI and Cloud Computing. Experienced in optical and SAR satellite image processing, land-cover mapping, change detection, agricultural monitoring and the development of scalable cloud-based geospatial workflows using Python, JavaScript, Google Earth Engine, Google Cloud, ArcGIS, QGIS and PostgreSQL/PostGIS. Strong background in applying Machine Learning and Deep Learning to large-scale environmental, agricultural and territorial monitoring projects for public institutions and international initiatives.

PROFESSIONAL EXPERIENCE

Optimization Programmer – EO & Land Cover, FUNDACIÓN NATURA Jun 2023 – Dec 2025

- Developed and evaluated automated and cloud-enabled methodologies for producing Colombia's national land-cover map at 1:100,000 scale, integrating Earth Observation data, geospatial processing and Artificial Intelligence.
- Designed and tested workflows to improve land-cover interpretation, processing efficiency and automation using new geospatial technologies and data sources.

Remote Sensing & Geospatial Consultant, 2020 – 2025 | Multiple consulting engagements UNIDAD DE PLANIFICACIÓN RURAL AGROPECUARIA – UPRA

- Developed, implemented and validated Machine Learning and cloud-based geospatial workflows for satellite-based crop mapping and agricultural monitoring at national scale.
- Automated satellite image processing using Google Earth Engine and cloud geospatial infrastructure to support Colombia's Municipal Agricultural Assessment System.

Satellite Imagery Professional, Sept 2019 – Jun 2025 AGENCIA NACIONAL DE MINERÍA – ANM

- Processed, interpreted and classified optical and SAR satellite imagery to support monitoring and oversight of mining titles across Colombia.
- Developed Python and JavaScript scripts to automate raster and vector geoprocessing, improving repeatability and efficiency of satellite-image analysis workflows.
- Applied remote sensing and image-processing methodologies to generate geospatial information supporting technical inspection and monitoring activities.

Remote Sensing Professional, Jul 2022 – Jun 2023 FIDUAGRARIA S.A. – BIOCARBONO ORINOQUÍA

- Supported the technical implementation of the project "Data infrastructure for the Colombian Orinoquía region and monitoring of crops and pastures for livestock"
- Applied satellite Remote Sensing, Machine Learning and cloud-based geospatial technologies to identify and monitor crops in support of rural land planning and low-carbon production.

Geoanalytics Engineer, OLGOONIK – U.S. EMBASSY Jul 2021 – May 2022

- Automated and optimized raster and vector geospatial analysis workflows for Colombia's illicit-crop monitoring system.

- Implemented and evaluated semi-automated algorithms for illicit-crop detection and monitoring in support of the Colombian National Police and the U.S. Embassy.

**Geospatial Data Processing & Analysis Professional,
INSTITUTO GEOGRÁFICO AGUSTÍN CODAZZI – IGAC**

Sept 2020 – Jul 2021

- Developed Python/ArcGIS scripts and toolboxes to automate raster and vector geoprocessing for cartographic production, processing and quality-control workflows.
- Developed and maintained Google Earth Engine applications for large-scale land-cover classification and change detection using cloud-based processing and Machine Learning workflows.
- Published and managed geographic services through ArcGIS Server/Online and OGC standards, and supported the organization and administration of geospatial datasets.

GIS Professional, UT GESTIÓN INMOBILIARIA

Jun 2018 – Sep 2019

- Designed, developed and managed a GIS application for the administration of Ecopetrol's real-estate rights and associated spatial information.
- Automated spatial and alphanumeric workflows using PHP and JavaScript and modelled and administered the application's PostgreSQL database.

**Geospatial Engineer – Defense & Security Sector,
PROCÁLCULO PROSIS S.A.**

Apr 2017 – Jul 2018

- Managed and supported the distribution and technical use of PlanetScope, DigitalGlobe and RADARSAT satellite imagery for Defense and Security organizations.
- Provided satellite image processing and geoprocessing support using ArcGIS, ERDAS, SOCET and PCI, and delivered technical training to end users.

**Assistant Geomatics Engineer – Exploration & Geosciences,
FRONTERA ENERGY CORP.**

Jul 2016 – Feb 2017

- Supported GIS, cartographic production and processing of optical and radar satellite imagery for land-cover mapping and geoscience applications.
- Managed raster/vector datasets, geodatabases and environmental-compliance geospatial deliverables.

EDUCATION

Erasmus Mundus MSc in Geospatial Technologies

Sept 2025 – Present

Universitat Jaume I, Spain · University of Münster, Germany

MSc in Information and Communication Sciences – Geomatics

Jan 2020 – Jun 2023

Universidad Distrital Francisco José de Caldas, Colombia

Specialization in Geographic Information Systems

Jan 2018 – Jun 2019

Universidad Distrital Francisco José de Caldas, Colombia

BSc in Cadastral and Geodesy Engineering

Feb 2012 – Jun 2017

Universidad Distrital Francisco José de Caldas, Colombia

TECHNICAL EXPERTISE

Earth Observation & Remote Sensing

Optical & SAR Imagery · Land-Cover Mapping · Change Detection · Satellite Image Time Series
· Agricultural Monitoring

Cloud & Geospatial Computing

Google Earth Engine · Google Cloud · Cloud-Based Geospatial Processing · Large-Scale EO Workflows

Programming & Spatial Data

Python · JavaScript · PostgreSQL/PostGIS · Raster & Vector Processing · Geospatial Automation

GIS & AI

ArcGIS Pro · QGIS · Machine Learning · Deep Learning · Spatial Analysis

PUBLICATIONS

Article "Artificial Intelligence and Cloud Computing for a New Generation of Corine Land Cover Maps in Colombia". Remote Sensing, 2026, 18, 1448. DOI: 10.3390/rs18101448

Article "Google Earth Engine App Using Sentinel-1 SAR and Deep Learning for Ocean Seep Methane Detection and Monitoring". Remote Sensing Applications: Society and Environment, 2023, 32, 101036. DOI: 10.1016/j.rsase.2023.101036.

Article "Change Detection over Mining Areas Using Deep Neural Networks from PlanetScope Image Time Series". IEEE Latin America Transactions, 2022, 20(9), 2196–2205.

Article "Evaluation of Classification Algorithms on the Google Earth Engine Platform for the Identification and Detection of Changes in Buildings in Rural Areas and Municipal Urban Perimeters Based on High-Resolution Images". Revista de Teledetección, 2021, 58, 71–88.

Article "Classification of Land Cover through Machine Learning Algorithms for Fusion of Sentinel-2A and PlanetScope Imagery". IEEE LAGIRS, 2020, 246–253. DOI: 10.1109/LAGIRS48042.2020.9165632.

Article "Generation of change data of land cover in the Bogotá savannah using time series with Landsat images and MODIS-Landsat synthetic images between 2007 and 2013". Revista de Teledetección, [S.l.], no. 54, pp. 41–58, Dec. 2019. ISSN 1988–8740.

Book: "Application for analyzing time series of satellite images." Editorial Académica Española, 2020. ISBN: 978–620–2–81018–0.

Book: "Time series and synthetic images for land cover changes." Editorial Académica Española, 2017. ISBN: 978–620–2–23008–7.

TEACHING EXPERIENCE

Universidad del Rosario / Ministry of Transport

Jul 2025

Intermediate & Advanced Geoprocessing in QGIS — Instructor | 20 hours

- Delivered training in advanced raster/vector spatial analysis, network analysis and spatial database management using open-source GIS tools.

LANGUAGES

Spanish — Native

English — B2 (CEFR)

English certification/training: New College Group, Liverpool, United Kingdom